



Class 10 Computer Science: Database Management System (DBMS)

The Complete, Recording-Ready Masterclass

CSITMASTERYHUB | CURRENT SYLLABUS



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Exam Definition - Data:

A collection of raw facts, figures, or information that may not have meaning on its own.

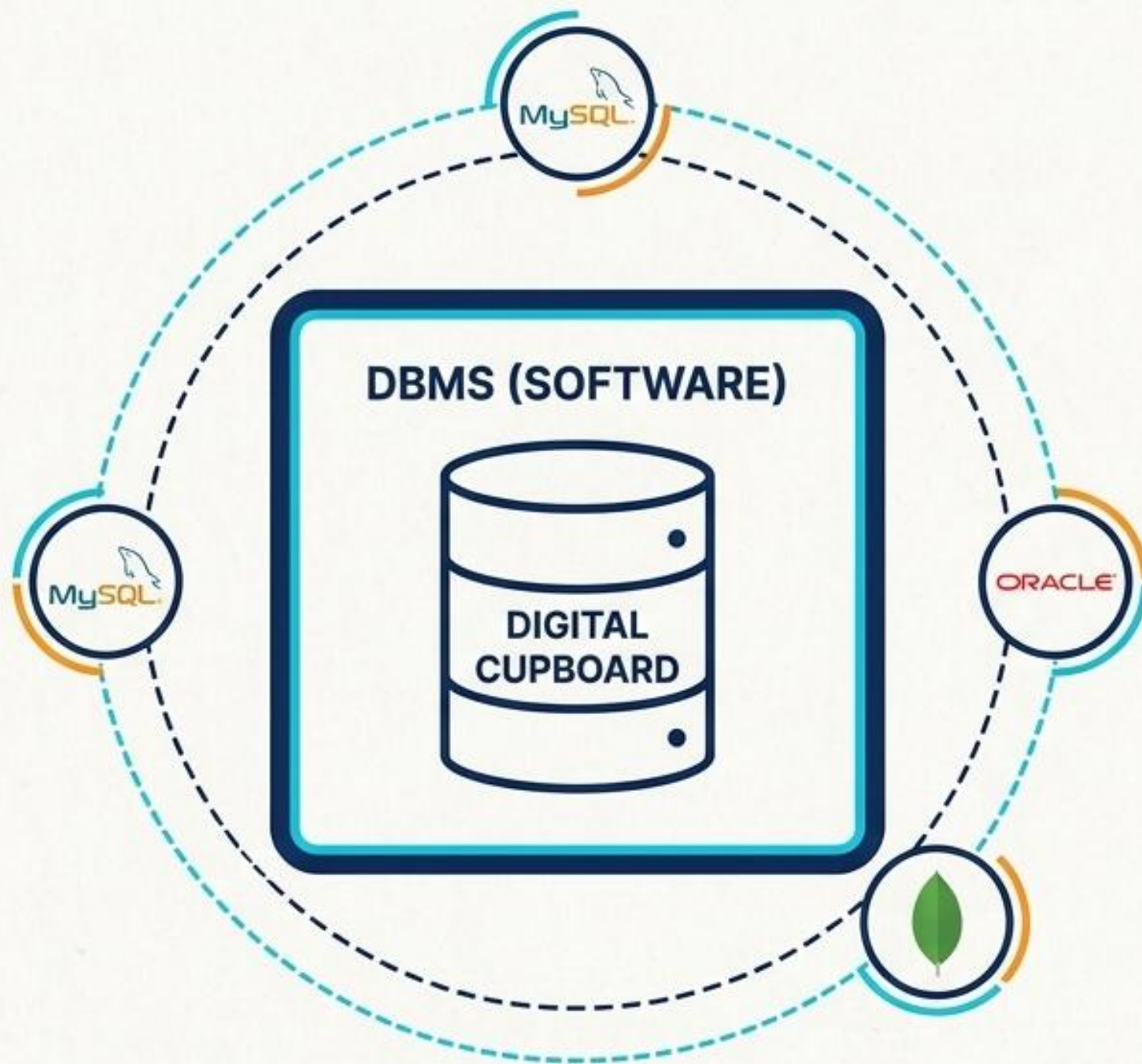


Exam Definition - Information:

The useful result we get after arranging or processing data.

Aspect	Data	Information
Definition	Raw, unprocessed facts or figure	Processed and organized data with meaning.
Meaning	No clear meaning on its own.	Meaningful and is useful for understanding or decision-making.
Example	2082, 95, Binay	Binay scored 95 marks in 2082 BS, and is one of the top performers.





Common Confusion: A Database is where the data lives. The DBMS is the software you use to talk to it.



Exam Definition - Database

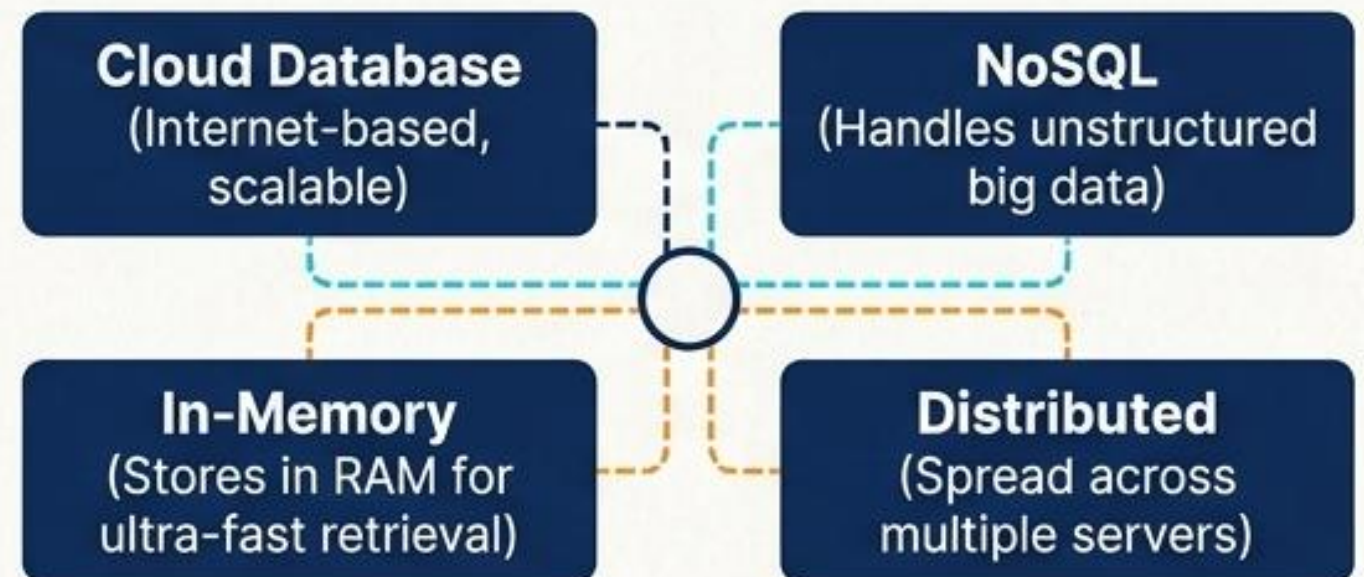
An organized collection of data that can be easily accessed, managed, and updated (stores info in tables).



Exam Definition - DBMS

Application software designed to store, manage, and organize data efficiently (acts as an intermediary between users and the database).

Recent Trends

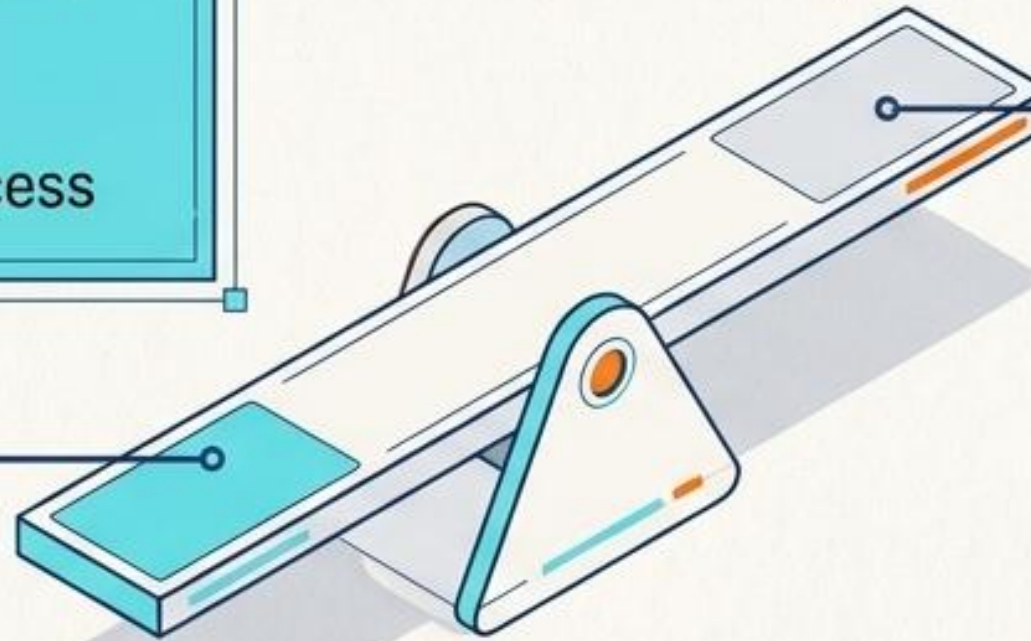


THE BENEFITS

- Data Integrity (consistent rules)
- Redundancy Control (minimizes duplication)
- Easy Retrieval & Storage
- Security & Multiple User Access

THE CHALLENGES

- High Initial Setup Cost
- Requires Technical Knowledge
- Complex Maintenance



★ Important for SEE: Why use a DBMS rather than a simple spreadsheet?

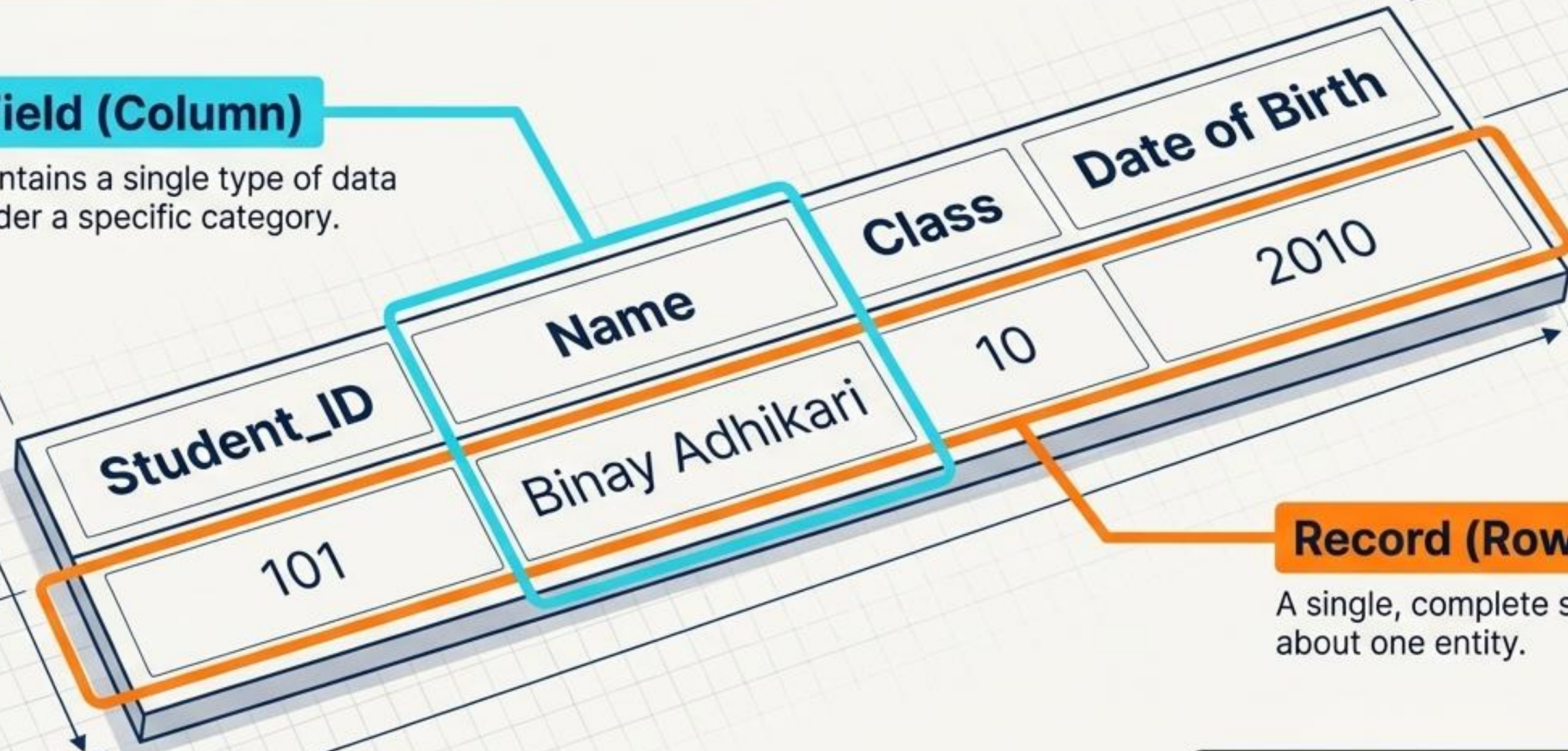


A DBMS is necessary to handle large amounts of data because it provides critical data redundancy control, ensures data consistency, and offers advanced security and multi-user access that simple spreadsheets cannot provide.

 **Exam Definition - Relational Table:** A structured way to store data in rows and columns where each table represents a specific entity and can connect to others.

Field (Column)

Contains a single type of data under a specific category.



Student_ID	Name	Class	Date of Birth
101	Binay Adhikari	10	2010

Record (Row)

A single, complete set of data about one entity.



Common Confusion:

Remember: Field = Column (Vertical),
Record = Row (Horizontal).



Primary Key

Exam Definition

Uniquely identifies each record.

Rules:

No duplicates allowed, cannot be NULL.
Only **ONE** per table.

Example:

StudentID (101, 102).



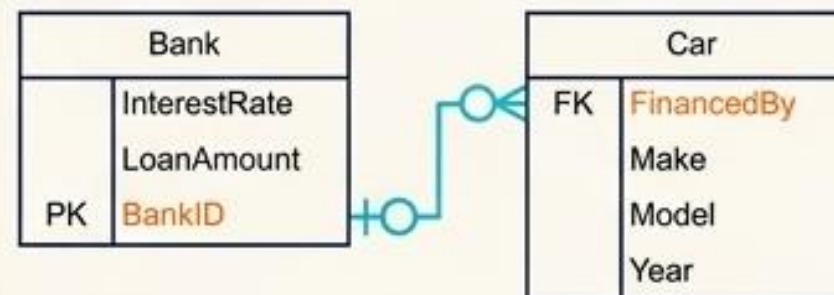
Foreign Key

Exam Definition

A field in one table that refers to the primary key in another table.

Purpose:

The 'linking pin' that enforces relationships and data integrity.



Composite Key

Exam Definition

Combining **two or more** fields to uniquely identify a record when a single field cannot.

Example:

StudentID + Subject

★ Important for SEE: Be ready to identify Primary and Foreign keys from a given table diagram!



 **Exam Definition - Relationship:** A logical connection between two or more tables, typically based on a common field (like a key).

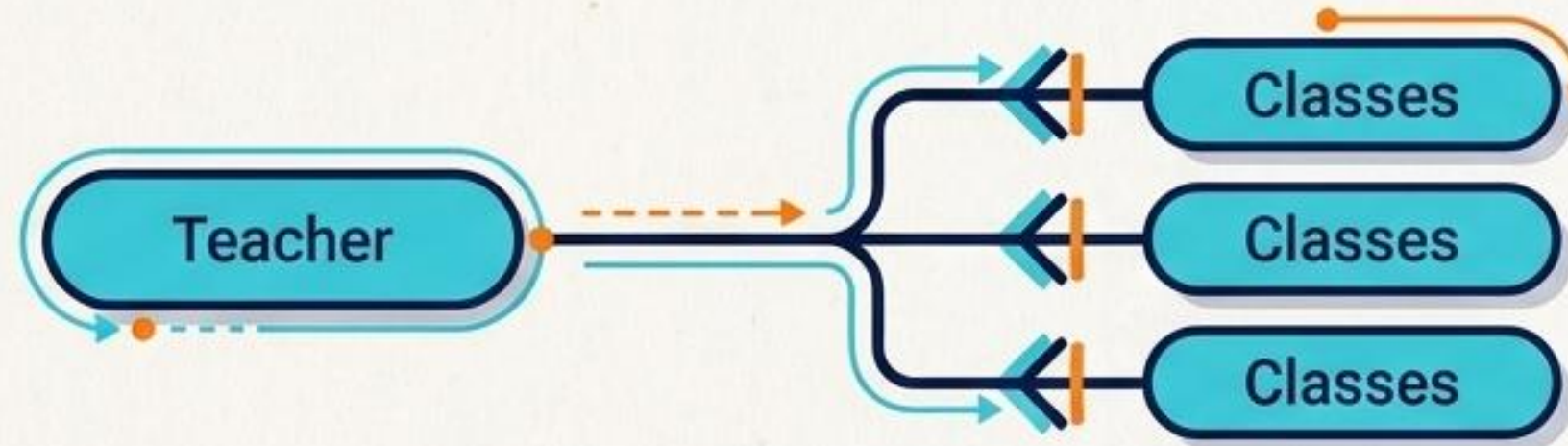
1:1 (One-to-One)

Each record connects to only one record.



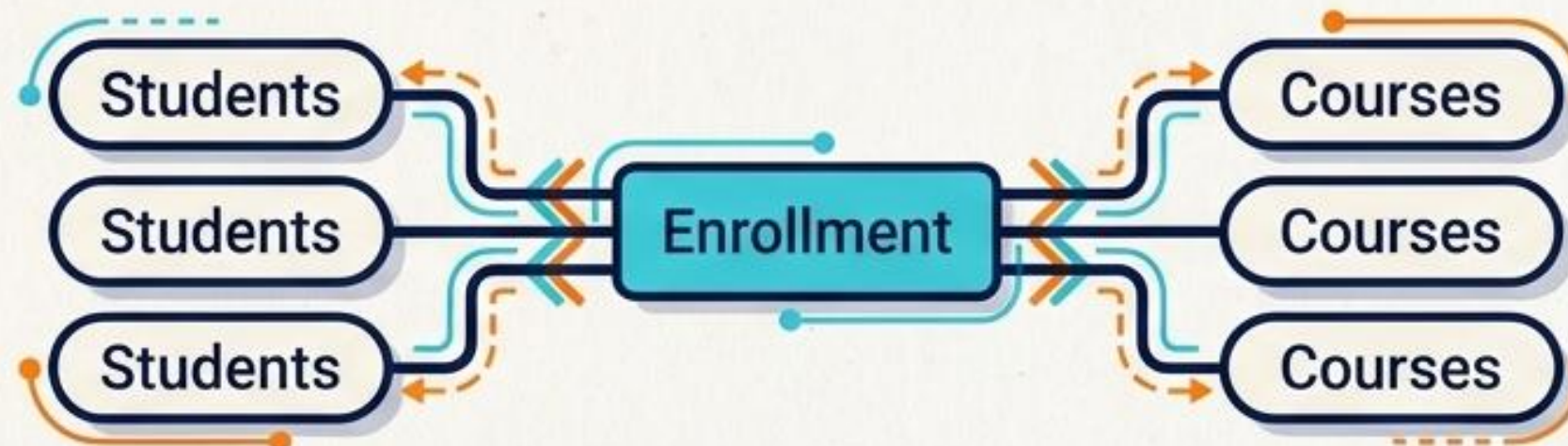
1:M (One-to-Many)

A single record links to multiple records in another table. (Most common).



M:M (Many-to-Many)

Requires a third "junction table" to link multiple records on both sides.



1. Numeric

INT (Whole numbers),
DECIMAL (Exact with decimals).

2. Date/Time

DATE (YYYY-MM-DD),
DATETIME (Date + Time).

3. Boolean

BOOLEAN (TRUE/FALSE).

4. Character (Text)

CHAR and **VARCHAR**.

★ Important for SEE: CHAR vs VARCHAR

CHAR(n)

- Fixed-length text.
- Always stores exactly n characters (pads with empty spaces).

VARCHAR(n)

- Variable-length text.
- Stores up to n characters. More flexible, saves space.

⚠ **Common Confusion:**
Don't use **INT** for phone numbers (they can have leading zeros or symbols). Use **VARCHAR**!



What is SQL?

DDL (Data Definition Language)

The Builder

Defines and manages the structure of the database (tables, schemas).

Key Trait: Automatically saves (commits); cannot be rolled back.

CREATE, ALTER, DROP, TRUNCATE

DML (Data Manipulation Language)

The Operator

Works directly with the content/records inside the tables.

Key Trait: Operations can be rolled back (unless committed).

INSERT, SELECT, UPDATE, DELETE

★ **Important for SEE:** Differentiating DDL vs DML is a highly tested concept!
Know which commands belong to which pillar.

DDL in Action: Building the Architecture

CREATE (Make it)

```
CREATE DATABASE mySchool;  
  
CREATE TABLE students (ID INT,  
Name VARCHAR(50));
```

DROP (Destroy it completely)

```
DROP DATABASE mySchool;  
DROP TABLE students;
```

TRUNCATE & ALTER (Modify it)

Truncate: Removes all records but keeps the empty table structure.

```
TRUNCATE TABLE students;
```

Alter: Changes table structure (e.g., adding a column).

```
ALTER TABLE students ADD Age INT;
```



Common Confusion: **DROP** deletes the entire table structure. **TRUNCATE** just empties the data inside it but leaves the table intact.

DML in Action: Manipulating the Content

INSERT (Add records)

```
INSERT INTO students(ID, Name)  
VALUES (1, 'Dikshya');
```

SELECT (View records)

```
SELECT * FROM students;
```

(The * means ALL columns).

UPDATE (Change records)

```
UPDATE students SET Class = 11  
WHERE ID = 5;
```

DELETE (Remove records)

```
DELETE FROM students  
WHERE Name = 'Muna';
```

★ **Important for SEE:** Always use a **WHERE clause** with **UPDATE** and **DELETE**! If you forget it, you will update/delete **EVERY** record in the entire table!

Precision Filtering

The **WHERE** Clause

Filters records based on specific conditions.

WHERE Class = 10

The **LIKE** Clause (Pattern Matching)

Searches for patterns using wildcards.

The % wildcard = zero or more characters.

LIKE 'A%' (Finds all names starting with A)

The _ wildcard = exactly one character.

LIKE '_man' (Finds Aman, Iman, etc.)



Exam Definition: Rules applied to columns to ensure valid, consistent data.

MySQL Constraints (The Security Gates)



PRIMARY KEY



FOREIGN KEY



NOT NULL

(Prevents empty values)



UNIQUE

(Prevents duplicates)



CHECK

(Enforces specific numerical or text conditions)



What is a Report?



Exam Definition - Report:
The organized presentation of data retrieved from one or more tables using SQL queries.

Summary: Total values (sales, student count).

Grouped: Sorted by category (by class or city).

Detail: Full specific records.

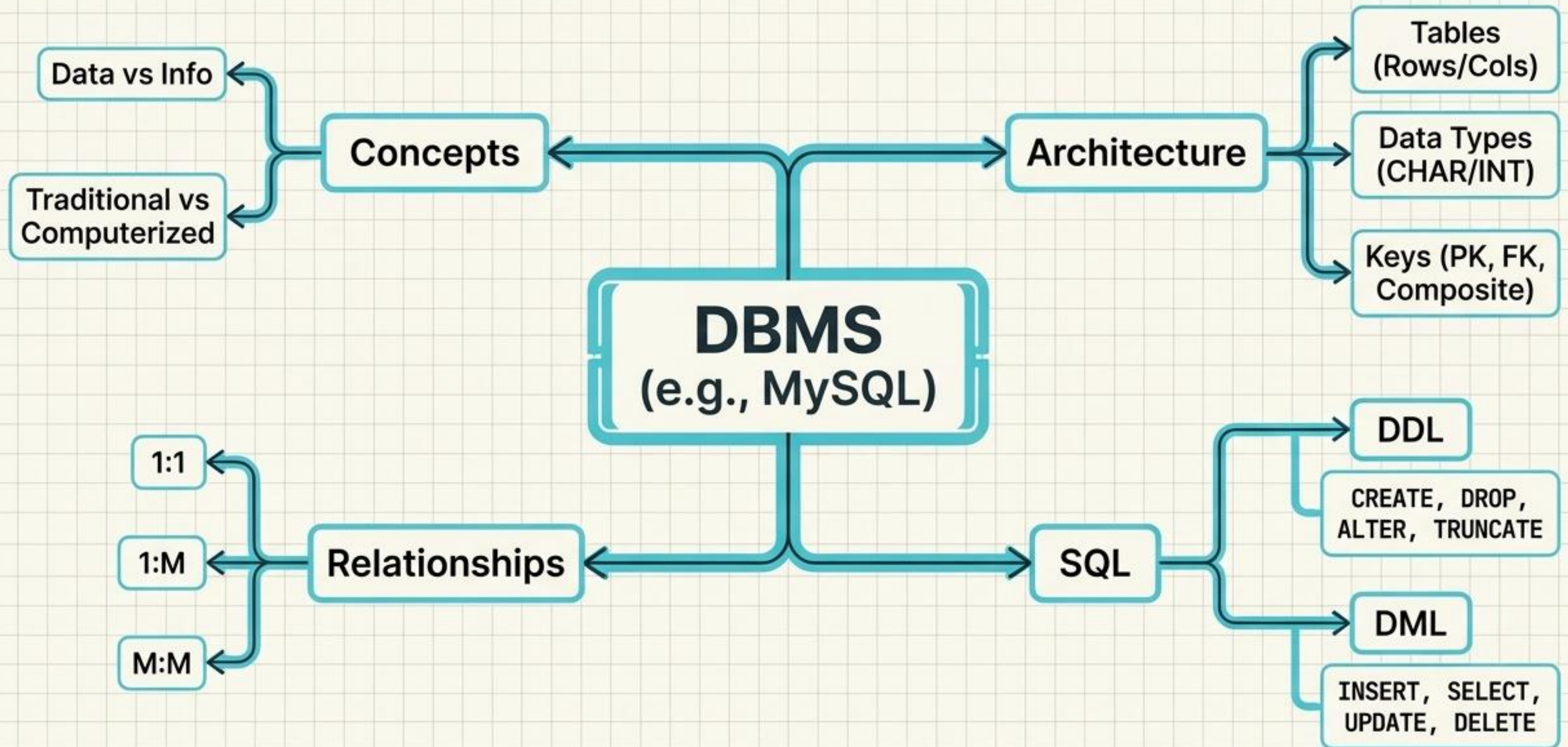
Filtered: Based on strict conditions (marks > 80%).



★ **Important for SEE:** Know the features of a report: Organized presentation, Data Filtering, Summarization, and Easy Export.



The 1-Minute Mindmap (Quick Revision Blueprint)



SEE Master Questions Checklist

★ Master these syllabus-aligned questions to guarantee top marks in the SEE Computer Science Exam!

Definitions to Master

- ☐ What is a database and DBMS?
- ☐ What is a primary key?
List its features.
- ☐ Define fields and rows.
- ☐ What is a foreign key?

Differentiate Between

- ☐ Data vs. Information.
- ☐ DDL vs. DML.
- ☐ CHAR vs. VARCHAR.
- ☐ Traditional File System vs. DBMS.

Conceptual & Applied

- ☐ Why is a DBMS better than a spreadsheet?
- ☐ Explain a one-to-many relationship with an example.
- ☐ What does the SQL LIKE clause help you do?
- ☐ What are SQL constraints and why are they important?

